Electron-Impact Excitation out of the Metastable Levels of Krypton

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We have measured the electron-impact excitation cross sections out of the two metastable levels of Kr into the ten levels of the $4p^5 5p$ configuration. For a common $4p^5 5p$ final level, the peak excitation cross sections out of the two individual $4p^5 5s$ metastable levels are found to differ by 1 to 2 orders of magnitude. This is explained by the special features of the electronic structure of the two configurations involved. The peak cross sections are 10 to 1600 times larger than the corresponding peak cross sections out of the ground state.

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Knowledge of electron-impact excitation cross sections of Kr out of both metastable levels is important intrinsically and for understanding gas discharges. Since ionization and excitation into higher levels from the metastable levels requires only low-energy electrons, many of these processes in gas discharges occur via two-step processes through the metastable levels. Furthermore, cross sections out of the metastable levels are much larger than cross sections from the ground state. Thus, the cross sections reported in this paper are of high interest for analysis of emissions from low temperature plasmas. For example, in a method known as trace rare-gas optical-emission spectroscopy, a trace amount of rare gases is added to a plasma and measurements of the radiation originating from emitting levels of the trace rare gases are used to determine the electron densities at different energies [1]. An emitting level may be populated by excitation out of the metastable levels with low-energy electrons ($\sim$2 eV) and by excitation out of the ground level with higher-energy electrons (>12 eV). Excitation cross sections for these processes are essential for obtaining the electron densities at a wide range of energies [2]. In this type of application, krypton offers advantages over Ar (and Ne) since the metastable levels of Kr are well separated in energy; the added Boltzmann factor from this energy difference leads to larger population differences between the two metastable levels than in Ar where the two metastable levels have a smaller energy separation. Moreover, the difference in the electronic structure between the heavy rare gases (Kr and Xe) and the lighter ones results in very different cross section behaviors which not only reveal fundamental connections between cross sections with atomic structure but also suggest simplifications in plasma applications.

The ground state of Kr is $4p^6\ {}^1S_0$. The lowest excited configuration of Kr is $4p^5 5s$. The $^2P_{3/2}$ and $^2P_{1/2}$ levels of the $4p^5$ ion core are separated by 0.6 eV. The $^2P_{3/2}$ ion core couples to the $5s$ electron to form $J = 2$ and $J = 1$ levels (referred to as the $1s_5$ and $1s_4$ levels, respectively, in Paschen's notation), and the $^2P_{1/2}$ ion core couples to the $5s$ electron to form $J = 1$ and $J = 0$ levels ($1s_2$ and $1s_3$). The $J = 2$ ($1s_5$) and $J = 0$ ($1s_3$) levels are metastable, with lifetimes on the orders of 0.5 s [3] and 30 s [4] for the $1s_3$ and $1s_5$ levels, respectively. The next excited configuration is $4p^5 5p$. The $^2P_{3/2}$ ion core couples to the $5p$ electron to produce six levels, denoted $2p_5$ through $2p_{10}$ ($J = 0, 2, 1, 2, 3, 1$), and the $^2P_{1/2}$ ion core couples to the $5p$ electron to produce four levels, $2p_1$ through $2p_4$ with $J = 0, 2, 1, 1$. We report measurements of the electron-excitation cross sections out of the $1s_3$ and $1s_5$ metastable levels into the ten $2p$ levels of the $4p^5 5p$ configuration.

Figure 1 shows the energy levels of each configuration in a two-tier structure resulting from the energy spacing of the $^2P_{3/2}$ and $^2P_{1/2}$ ion cores (0.6 eV). This spacing is comparable to the energy span of the $4p^5 5p$ levels with a given ion core which is a measure of the interaction of the core

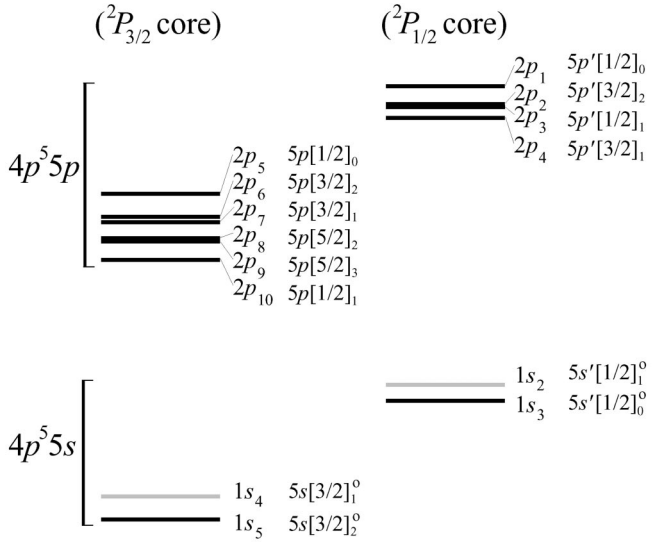


FIG. 1. Relevant energy levels of Kr labeled in both the Paschen and Racah notations. In Racah notation a level in the $4p^5 nl$ configuration is denoted as $nl[K]_J$ where K is the vector sum of l and J_c, the total angular momentum of the $4p^5$ core. Levels with $J_c = 1/2$ are denoted by a prime on the nl; levels with $J_c = 3/2$ are left unprimed.

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with the outer electron, and is also comparable to the energy separation between the highest levels in the $4p^5 5s$ configuration and the lowest level in the $4p^5 5p$. If the energy separation between the $^2P_{3/2}$ and $^2P_{1/2}$ core levels is much larger than the interaction of the core with the outer electron, one would expect the $5s \rightarrow 5p$ excitation to predominately preserve the core, otherwise a single collision would have to both excite the outer electron and change the core. Since, however, the interactions are comparable, as indicated by the energy level pattern, the core-changing excitations may be appreciable, and it is of great interest to determine the extent to which this occurs. In contrast, the smaller $^2P_{3/2}$ to $^2P_{1/2}$ core level spacing in Ar (0.17 eV) is overwhelmed by coupling of the Ar^+ core with the outer electron ($4p$) so that designation in terms of ion-core levels is not applicable to the Ar($3p^5 4p$) levels.

Figure 2 depicts the layout of the collision region in our apparatus. The metastable Kr atoms are produced in a hollow cathode discharge. A 1 mm diameter hole in the base of the hollow cathode generates a thermal atomic beam containing Kr atoms in both metastable levels and the ground level. The atomic beam is crossed at right angles by an electron beam. Fluorescence from the $2p$ levels is observed at right angles to the atomic beam and at 60° to the electron beam. Radiation from a particular $2p$ level is selected using a narrow bandwidth interference filter and is detected using a photomultiplier tube.

The thermal atomic beam is a mixed target containing atoms in both metastable levels and a large number of ground state atoms. At energies below the threshold for ground level excitation, the emission is solely due to excitation out of the two metastable levels. We determine the emission due to excitation out of a given metastable level by laser quenching. We use an Ar-ion pumped, single-mode, ring Ti:sapphire laser to pump atoms out of one of the metastable levels into a higher level. The laser beam crosses the atomic beam at the base of the hollow cathode discharge, before the atoms reach the electron-collision region. Atoms excited into the higher level can decay back to the metastable level, or to resonance levels that decay to the ground state. With greater than 200 mW of laser power, the target consists of only ground state atoms and atoms in the remaining metastable level. Measurements of the emission intensities from the various $2p$ levels yield the relative excitation cross sections into these levels out of the $1s_3$ and $1s_5$ levels.

Two additional experiments are needed for absolute measurements. The $1s_5 \rightarrow 2p_9$ cross section is measured with our fast-beam apparatus which creates a metastable target using near resonant charge exchange between a 2.5 keV Kr^+ beam and a Cs vapor target as described in Ref. [5]. We obtain absolute results by using the signal from the well characterized metastable beam target, the signal from a static gas target of ground state Kr, and the known cross section for excitation out of the Kr ground state into the $2p_9$ level [6]. The second additional measurement is the ratio of the $1s_3$ to $1s_5$ number densities in the hollow cathode discharge target. The $1s_3$ to $1s_5$ ratio is measured using laser induced fluorescence in a manner similar to that described in Ref. [7]. In this case, the laser beam from the Ti:sapphire laser intersects the atomic beam

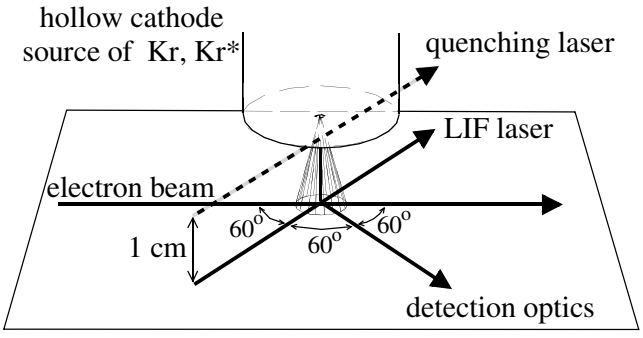


FIG. 2. Collision region of the hollow cathode source of metastable Kr atoms.

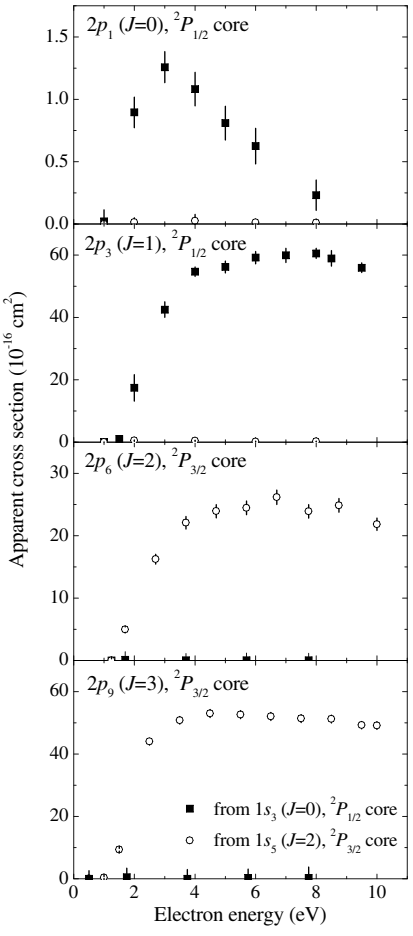


FIG. 3. Cross sections for excitation into the $2p_1$, $2p_3$, $2p_6$, and $2p_9$ levels from the $1s_3$ and $1s_5$ levels. Error bars are statistical only.

axis at right angles and in the same plane as the electron beam and the fluorescence detection axis.

The energy dependence of the excitation cross sections for four of the $4p^5 5p$ levels is shown in Fig. 3. Each plot includes the separately measured cross sections from each metastable level. To illustrate the full range of interesting phenomena, we discuss the results in terms of both the J value of the final level and the ion core associated with the level. As illustrated by the results in Fig. 3 for the $2p_6$ and $2p_9$ levels, the cross sections for excitation into the six $2p$ levels with the $^2P_{3/2}$ ion core have large cross sections out of the $1s_5$ metastable level which is also associated with the $^2P_{3/2}$ ion core (core-preserving excitation). On the contrary, the core-changing cross sections into these levels from the $1s_3$ metastable level ($^2P_{1/2}$ ion core) are very small, thus confirming the inhibition of core-changing excitation. (It is difficult to derive precise values for these cross sections due to the large statistical uncertainties introduced by subtracting two signal rates with the quenching laser on and off.) For excitation into the four $2p$ levels with the $^2P_{1/2}$ ion core, the roles are reversed with large cross sections out of the $1s_3$ metastable level. Since the $1s_5$ to $1s_3$ ratio of atoms in the source is about 12, the $1s_5$ contribution to the fluorescence signal rate is non-negligible even though the cross sections are small. Since the change in signal rate with the laser quenching on and off is now greater than the statistical uncertainties, we can also obtain core-changing cross sections for some levels.

The shapes of the excitation cross sections as a function of electron energy fall into two main categories. For core-preserving, dipole-allowed excitations ($\Delta J = 0, \pm 1, 0 \not\to 0$), there is broad energy dependence with a peak around 6 eV. The other shape is for the core-preserving, dipole-forbidden excitations [$1s_3(J = 0) \to 2p_1(J = 0)$, $1s_5(J = 2) \to 2p_5(J = 0)$], and is a sharp peak at 3 eV, decreasing by about a factor of 10 by 10 eV. Excitation into the $J = 2$ $2p_2$ level is an interesting case. Excitation into this level is core-preserving, dipole-forbidden from the $1s_3$ level, and core-changing, dipole-allowed from the $1s_5$ level. In Kr the core-preserving nature of collisions is paramount as illustrated by the results from this level shown in Fig. 4. The shape of the $1s_3 \to 2p_2$ excitation function is the sharp peak expected of a core-preserving, dipole-forbidden excitation. What is surprising is that the $1s_5 \to 2p_2$ excitation function also exhibits a low-energy peak which is not expected for dipole-allowed excitation. Furthermore, the cross section for dipole-forbidden excitation out of the $1s_3$ level is about 10 times larger than the dipole-allowed excitation out of the $1s_5$ level. Here the dipole-type coupling plays a secondary role in deference to core considerations.

Using our charge-exchange fast-beam apparatus, we have measured cross sections for excitation into the $2p_6$ and $2p_9$ levels out of the $1s_5$ metastable level for electron energies up to 400 eV. Based upon the Born-Bethe theory we extract, from these cross sections, values for the $1s_5 - 2p_6$ and $1s_5 - 2p_9$ optical oscillator strengths which agree with optical measurements to within 20% [8]. Although the Bethe theory is a high energy approximation, we found the dipole-type cross sections to track with the corresponding optical oscillator strengths even at very low energies except when the oscillator strength is very small (or zero). Here the coupling of initial to final state is no longer dominated by the dipolelike interaction with the incident electron, hence the cross sections are larger than expected from oscillator strengths alone. In Table I we list our measured

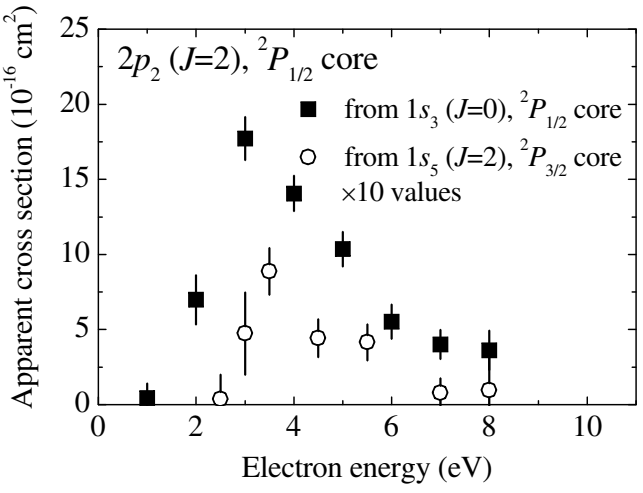


FIG. 4. Cross section values for excitation into the $J = 2$ $2p_2$ level. Note that the values for the $1s_5 \to 2p_2$ cross section have been multiplied by ten.

TABLE I. Comparison of the measured $1s_y \to 2p_x$ cross sections (in 10^{-16} cm^2) at 3.5 eV with the corresponding optical oscillator strengths f_{ij} multiplied by a normalization factor, C, chosen to match the $1s_5 \to 2p_9$ cross section. Error bars include both the statistical uncertainties and the ±35% uncertainties in the absolute calibration of the $2p_9$ level.

Excitation $i \to j$	Q (3.5 eV) (10^{-16} cm^2)	$f_{ij} \times C$ (scaled)
$1s_3 \to 2p_1$	1.3 ± 0.5	0
$1s_3 \to 2p_2$	16 ± 6	0
$1s_5 \to 2p_2$	0.9 ± 0.4	0.05
$1s_3 \to 2p_3$	55 ± 21	59
$1s_5 \to 2p_3$	0.5 ± 0.3	0.28
$1s_3 \to 2p_4$	64 ± 25	48
$1s_5 \to 2p_5$	0.4 ± 0.2	0
$1s_5 \to 2p_6$	21 ± 7	25
$1s_5 \to 2p_7$	3.2 ± 1.1	2.4
$1s_5 \to 2p_8$	8.9 ± 3.1	9.2
$1s_5 \to 2p_9$	52 ± 18	(52)
$1s_5 \to 2p_{10}$	15 ± 7	17

cross sections at 3.5 eV compared to the relative oscillator strengths scaled to match the $1s_5$-to-$2p_9$ cross section.

Hyman [9] applied the Born approximation to calculate the $4p^5 5s \rightarrow 4p^5 5p$ excitation cross sections but presented only cross sections that are averaged over the two configurations. Recently Dasgupta *et al.* [10] reported calculations using two different distorted-wave calculations (DW-1 and DW-2) and the results from using Breit-Pauli *R*-matrix close-coupling method with 15 basis functions (BP15) and with 51 basis functions (BP51). The *R*-matrix calculation with the smaller basis set (BP15) agrees better with our experiment than does the one with the larger set (BP51). Results from the two *R*-matrix calculations sometimes differ by a factor of 2 which points to potential problems with convergence. Theoretical calculations of these cross sections are challenging because of the complexity of the atomic structure of Kr and the complexity of the coupling of orbital and spin angular momenta of the outer electron with those of the core. Nonetheless, the ratio of the average cross section results for excitation from the $1s_5$ level for this experiment to the theoretical results of the BP15 and DW-1 calculations are 1.3 and 1.1, which is within the experimental uncertainties. Theoretical values for excitation cross sections out of the $1s_3$ level are generally smaller than experiment by about 40% for both the BP15 and DW-1 calculations, which is just at the limit of the experimental uncertainties. The values from the DW-2 calculation, which were not unitarized, are generally too large below 30 eV.

Our reported apparent cross sections include both direct electron excitation into the level of interest and an additional cascade contribution from excitation into higher levels that decay into the level of interest. We have previously found that for dipole-allowed metastable excitation processes the cascade contributions are less than 10% [11]. Based on this, we estimate that cascade corrections will reduce our reported cross sections by less than 10% for most of the levels.

Previous measurements of some of the reported cross sections have been made by Mityureva, Penkin, and Smirnov [12]. We find the same energy dependence of the cross sections for the four levels ($2p_6$, $2p_7$, $2p_8$, and $2p_9$) published in Ref. [12]. However, the magnitude of their cross sections are between a factor of 10 to 20 times larger than our values. More recently, Kolokolov and Terekhova [13] have deduced the metastable excitation cross sections from afterglow studies of a He-Kr plasma. Their rate coefficients yield cross sections that are generally 4 to 5 times larger than our results.

Our results are the first reliable experimental determination of the electron-excitation cross sections out of the metastable levels of Kr. The peak excitation cross sections into a particular excited level from a metastable level are 10 to 1550 times larger than corresponding peak cross sections out of the ground state [6]. We find that the two-tier structure of Kr energy levels plays a big role in determining the sizes of excitation cross sections, with core-changing cross sections being 1 to 2 orders of magnitude smaller than the corresponding core-preserving ones. Thus in plasma diagnostic applications, excitations into some of the $2p$ levels out of one metastable level are so much stronger than the excitations out of the ground level and the other metastable level that the analysis of the electron energy distribution involves only one metastable level making the analysis much simpler than when all three levels are involved. This greatly enhances understanding and insight into the processes involved which are crucial to the analysis of low temperature plasmas.

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